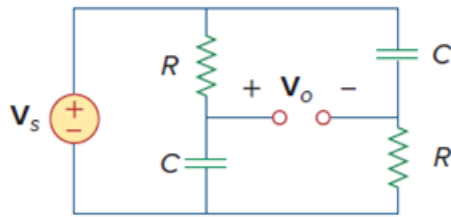


Problem

For the phase shifter circuit shown in Fig. 14.107, find $H = V_o/V_s$.

Figure 14.107:



Step-by-step solution

Step 1 of 6

Consider the following phase shifter circuit diagram:

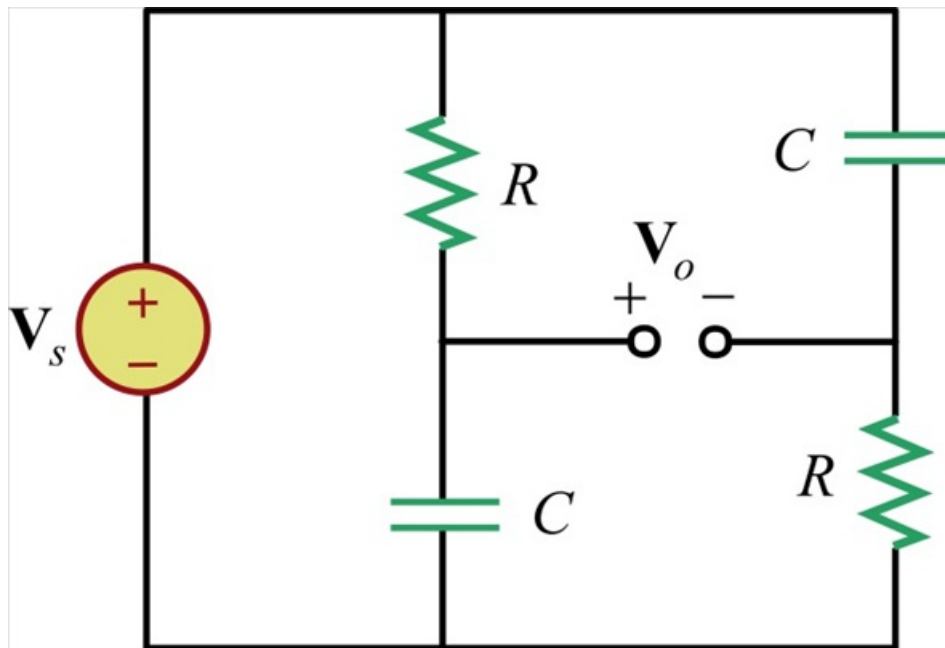


Figure 1: Phase shifter.

Step 2 of 6

Draw the following circuit diagram to find voltage V_{o+} .

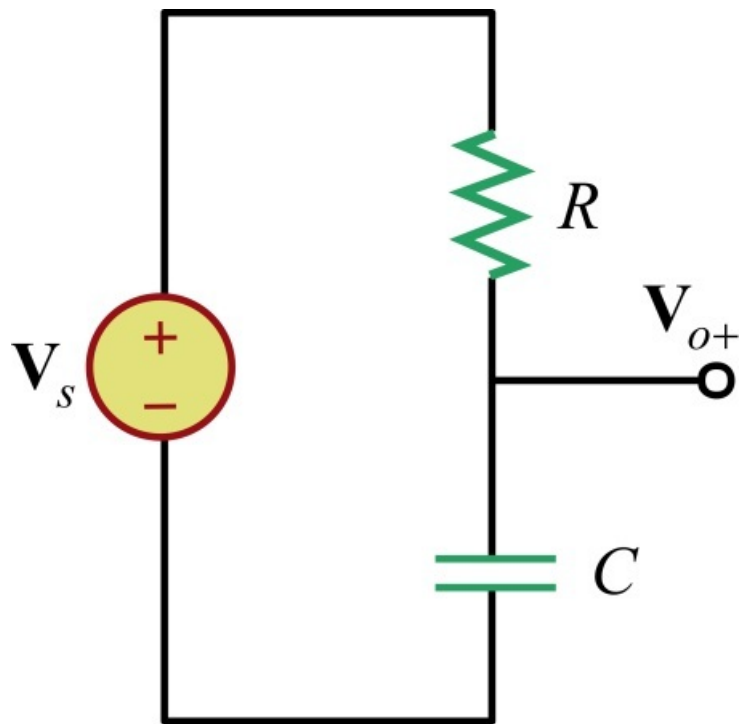


Figure 2

Step 3 of 6

Calculate the voltage V_{o+} .

Apply voltage division rule across the capacitor in Figure 2.

$$\begin{aligned} V_{o+} &= \frac{\left(\frac{1}{j\omega C} \right)}{R + \frac{1}{j\omega C}} V_s \\ &= \frac{1}{1 + j\omega RC} V_s \end{aligned}$$

Therefore, the value of voltage V_{o+} is $\frac{1}{1 + j\omega RC} V_s$.

Step 4 of 6

Draw the following circuit diagram to find voltage V_{o-} .

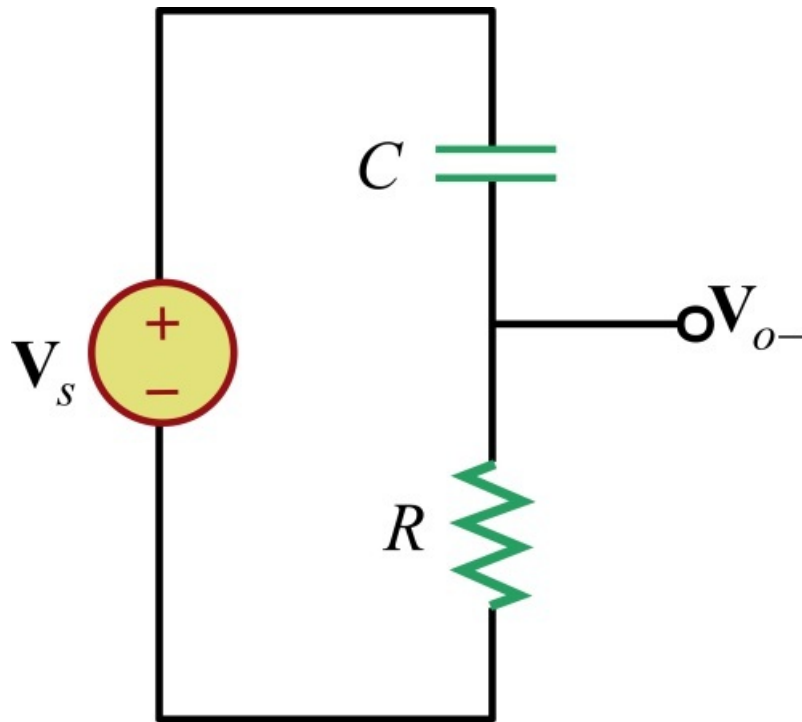


Figure 3

Step 5 of 6

Calculate the voltage V_{o-} .

Apply voltage division rule across the resistor in Figure 3.

$$\begin{aligned} V_{o-} &= \frac{R}{R + \frac{1}{j\omega C}} V_s \\ &= \frac{j\omega RC}{1 + j\omega RC} V_s \end{aligned}$$

Therefore, the value of voltage V_{o-} is $\frac{j\omega RC}{1 + j\omega RC} V_s$.

Step 6 of 6

Calculate the voltage V_o .

Write the expression for voltage V_o .

$$V_o = V_{o+} - V_{o-}$$

Substitute $\frac{1}{1+j\omega RC}V_s$ for V_{o+} , $\frac{j\omega RC}{1+j\omega RC}V_s$ for V_{o-} in this expression.

$$\begin{aligned} V_o &= \frac{1}{1+j\omega RC}V_s - \frac{j\omega RC}{1+j\omega RC}V_s \\ &= \frac{1-j\omega RC}{1+j\omega RC}V_s \end{aligned}$$

Therefore, the voltage V_o is $\frac{1-j\omega RC}{1+j\omega RC}V_s$.

Calculate the value of transfer function of phase shifter H .

Write the expression for H .

$$H = \frac{V_o}{V_s}$$

Substitute $\frac{1-j\omega RC}{1+j\omega RC}V_s$ for V_o in this expression.

$$\begin{aligned} H &= \left(\frac{1-j\omega RC}{1+j\omega RC} \right) \frac{V_s}{V_s} \\ &= \frac{1-j\omega RC}{1+j\omega RC} \end{aligned}$$

Replace s for $j\omega$ in this expression.

$$H = \frac{1-sRC}{1+sRC}$$

Therefore, the value of H is $\boxed{\frac{1-sRC}{1+sRC}}$.